MEGHANA RAJAKUMAR

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XR Engineer - Building Enterprise solutions with AI, XR, Computer Vision and Cloud Automation

Technical Project Lead and hands-on engineer with 6+ years of experience designing and deploying enterprise AI, computer vision, XR, and cloud solutions at BMW Group and Epsilon (Publicis Groupe). Experienced in vision-language model fine-tuning, agentic AI, full-stack development, Unity-based XR, and cloud automation. I work across design and engineering, taking ideas from Figma prototypes to production code, and validating them with real users along the way. Leads cross-functional initiatives from discovery and architecture through prototyping, deployment, and production, combining hands-on engineering with technical stakeholder and partner leadership.

CORE COMPETENCIES

Partner & Solution Leadership
- Partner Enablement & Technical Documentation
- Agile Project Delivery
- Solution Design & Architecture
- Reviews
- Stakeholder & Partner Management
- SDK & Developer Tooling Development
- Prototyping to Production Scaling
- Product Roadmap & Strategy

XR & UI/UX Design
- Enterprise XR Solution Architecture
- AR for Manufacturing & Training
- Smart Glasses & Wearable XR
- Hand Tracking, Mixed Reality (Passthrough)
- Performance Profiling
- Digital Twin Development & Simulation Design
- Real-Time 3D Pipelines: Modeling, Texturing, Rigging, Animation
- XR Performance Optimization
- UX & Interaction Design
- Rapid Prototyping (Figma, Figma Make)
- User Research & Usability Testing

AI, Physical AI & Cloud
- SLM & LLM Fine-Tuning for Physical AI
- Vision-Based AI Workflows
- Edge & On-Device AI Deployment
- Agentic AI Workflows
- Dataset Curation & Domain-Specific Model Training
- Cloud Automation (AWS, Azure), CI/CD & DevOps

Platforms & Tools: Unity, Unreal Engine, Meta Quest 3/Pro, Meta Horizon OS (Android APK pipeline), C++, C#, Claude Code, Cursor & Codex, Github Copilot, AWS, Azure, Terraform, Ansible, gRPC, FastAPI, and TypeScript (React & Angular), Figma, Figma Make

PROFESSIONAL EXPERIENCE

IT Technical Project Lead and Technical Artist

BMW Group – Greenville, South Carolina **Nov 2023 - Present**

- **Designed and built SmartEye Video Processor end to end as sole developer**, a self-service data preparation platform that centralizes video dataset collection, preprocessing, and standardization for AI model training across BMW developer teams. Developed the Python/FastAPI backend and Angular frontend using Density (BMW's design system) to segment uploaded video into dataset groups, transcribe and label content, and apply configurable transforms and anonymization to protect the identities of people on camera. Owned client demos, user support, handover, and industrialization.
- **Built 3D content, interaction systems, and the asset pipeline for Impossible Battery, a mixed reality digital twin of BMW's fifth-generation high-voltage EV battery for Meta Quest 3/Pro,** now used at BMW training centers in the US, Germany, South Africa, and China to train technicians on high-voltage battery repair. Modeled, textured, rigged, and animated 70+ assets, built the hand-tracked tool interaction system in Unity (C#), and built a Pixyz-to-Unity CAD pipeline that automates ~80% of asset optimization. Demoed to internal teams and partners including Meta; featured on Meta's blog and covered by UploadVR.
- **Owned Meta Quest 3/Pro testing and performance optimization**, profiling each build on both devices to catch bugs and interaction issues, and reducing APK size while holding 90 FPS to prevent motion sickness. Joined client feedback sessions after each test build and turned requested changes into the next iteration.
- **Fine-tuned and evaluated vision models for SmartEye Human Amplification, BMW's Vision AI system that automates a 160+ step vehicle quality checklist** previously done manually over days. Compared YOLO and vision-language models for video action detection, then labeled production-line footage and fine-tuned NVIDIA's NVILA to recognize technician actions, auto-check completed steps, and flag missed ones. Tested egocentric wearables (Meta Ray-Ban, Seeed Studio) against fixed-position cameras for data capture and assessed edge feasibility for real-time inference, enabling hands-free QA while reducing latency and inference cost.

- **Led discovery and built the frontend for an AI-driven Legal Defense platform** that consolidates hundreds of case documents into a single vehicle timeline and flags discrepancies and evidentiary gaps. Interviewed technicians and field analysts to map BMW's litigation review workflow as process SME, and built the prototype that demonstrated time savings on a real case for Gateway review. Developed the React frontend from UX designs, integrated backend APIs, coordinated design and backend teams, and ran beta testing. Now in pilot, with a conservative projected **ROI of $22M/yr.**
- **Contributed platform research, frontend optimization, and a capability framework to Customer Companion, BMW's agentic commerce app in ChatGPT**, now live in Europe. Tested the app's MCP servers and assessed app-building capabilities across AI chat platforms including Microsoft Copilot and Google Gemini. Applied traditional RAG, semantic search embeddings, and progressive profiling to maximize personalization across customer evaluation and purchase decisions.
- **Acted as the primary technical point of contact for a formal external partnership with SCADPro**, guiding an academic partner team through the full integration lifecycle of a live BMW XR program – from early concept and architecture review through prototype testing to production rollout – translating partner feedback into technical requirements and running iterative demonstrations on weekly feedback cycles.

Software Engineer II and Cloud Automation Expert

Epsilon (Publicis Groupe) – Bangalore, India **Jul 2019 - Sep 2022**

- Designed, implemented, and scaled infrastructure automation and deployment pipelines using Terraform, Ansible, and CI/CD frameworks, driving a 20% reduction in deployment times and a 32% improvement in overall deployment efficiency by eliminating manual processes.
- Led complex cloud and production operations, including specialized migrations across AWS and Azure, end-to-end orchestration of critical workflows and data transitions, and major production release upgrades, identifying risks, defining mitigation strategies, and minimizing downtime while improving system stability and reliability.
- Built internal dashboards and tools in Angular and React (TypeScript) to support deployment and production operations teams.
- Owned production reliability and team enablement, rapidly resolving high-impact issues across production tenants to ensure high availability and consistent performance, while also supporting hiring efforts by interviewing entry-level technical candidates.

EDUCATION

Master of Arts - *Magna Cum Laude (3.7 GPA)* **09/2022 – 06/2024**

Savannah College of Art & Design - Savannah, USA

- Major in Animation, Interactive Technology, Special Effects and Video Graphics

Bachelor of Engineering - *Summa Cum Laude (4.0 GPA)* **08/2015 – 07/2019**

RV College of Engineering - Bangalore, India

- Major in Computers and Telecommunication Engineering

PROFESSIONAL & ACADEMIC ACHIEVEMENTS AND AWARDS

- **Gold Medalist in Engineering**, awarded for outstanding academic excellence and top performance in the program
- **Interview Panelist** at the SCAD Faculty Conference, representing both BMW Group and SCAD; featured in the official SCAD Events Blog
- Published researcher with **411+ IEEE citations**, bridging applied research and real-world technology commercialization
- Crowned first place in the '**SHARK TANK**' – Epsilon game-changers competition among 300 teams from 10 + Nations